A Universal Turing Machine

A limitation of Turing Machines:

Turing Machines are "hardwired"

they execute only one program

Real Computers are re-programmable

Solution: Universal Turing Machine

Attributes:

- Reprogrammable machine

- Simulates any other Turing Machine

Universal Turing Machine

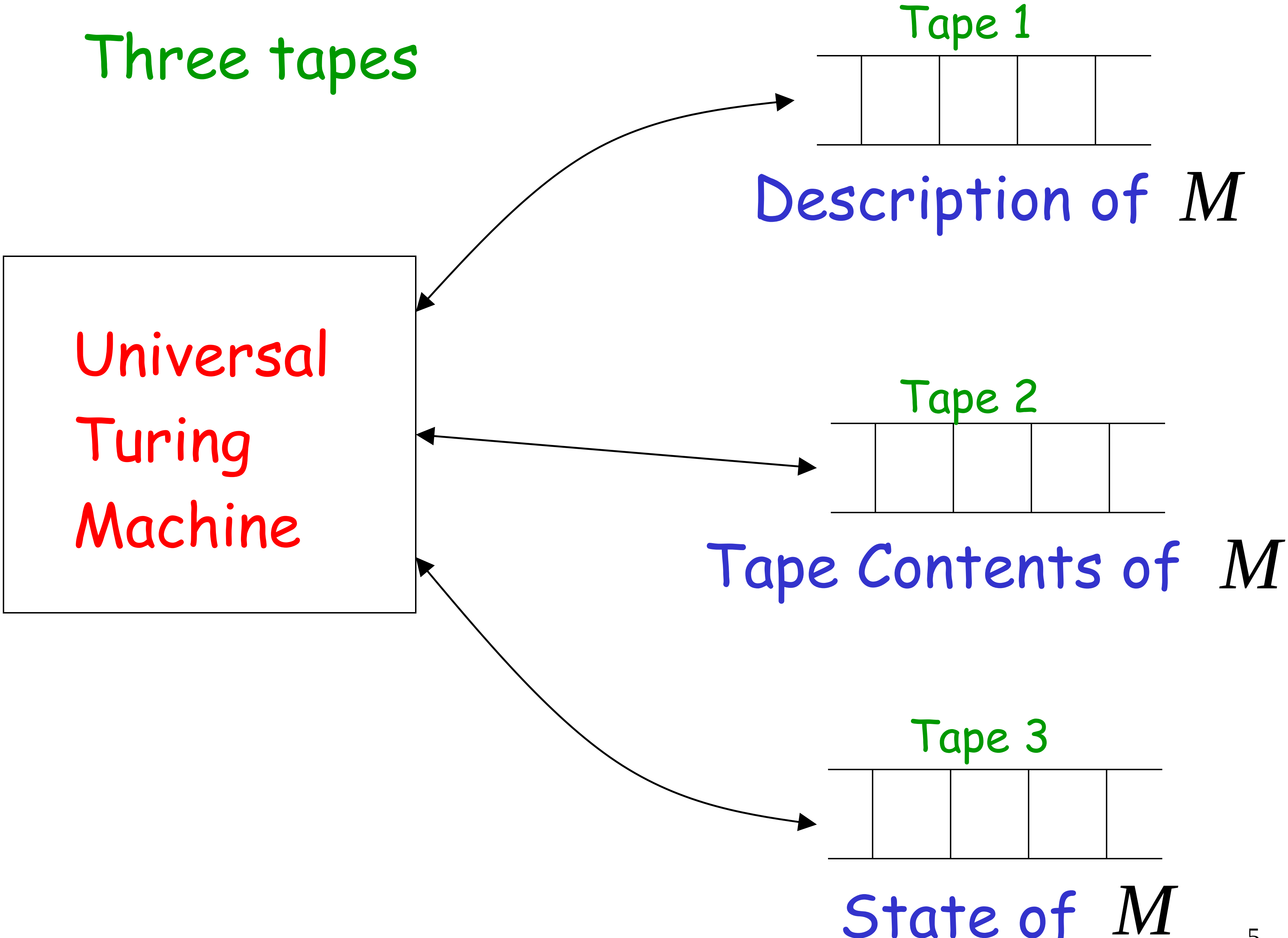
simulates any Turing Machine M

Input of Universal Turing Machine:

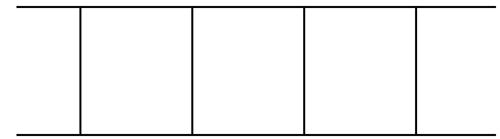
Description of transitions of M

Input string of M

Three tapes


Tape 1
Description of M
Universal
Turing
Machine
Tape 2
Tape Contents of M
Tape 3
State of M

Tape 1



Description of M

We describe Turing machine M
as a string of symbols:

We encode M as a string of symbols

Alphabet Encoding

Symbols:	a	b	c	d	$\cdots$
	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	
Encoding:	1	11	111	1111	

State Encoding

States:	q_1	q_2	q_3	q_4	$\cdots$
	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	
Encoding:	1	11	111	1111	

Head Move Encoding

Move:	L	R
	$\downarrow$	$\downarrow$
Encoding:	1	11

Transition Encoding

Transition: $\delta(q_1, a) = (q_2, b, L)$

Encoding: $1\,0\,1\,0\,11\,0\,11\,0\,1$

separator

Turing Machine Encoding

Transitions:

$$\delta(q_1, a) = (q_2, b, L) \qquad \delta(q_2, b) = (q_3, c, R)$$

Encoding:

$$10101101101 \; 00 \; 1101101110111011$$

separator

Tape 1 contents of Universal Turing Machine:

binary encoding
of the simulated machine M

Tape 1

1 0 1 0 11 0 11 0 10011 0 1 10 111 0 111 0 1100 …

A Turing Machine is described with a binary string of 0's and 1's

Therefore:

The set of Turing machines forms a language:

each string of this language is the binary encoding of a Turing Machine

Language of Turing Machines

L = { 010100101, (Turing Machine 1)

00100100101111, (Turing Machine 2)

111010011110010101, ...

...... }

Countable Sets

Infinite sets are either:

Countable

or

Uncountable

Countable set:

There is a one to one correspondence

of

elements of the set

to

Natural numbers (Positive Integers)

(every element of the set is mapped to a number
such that no two elements are mapped to same number)

Example: The set of even integers is countable

Even integers: (positive) $0, \; 2, \; 4, \; 6, \; \ldots$

Correspondence: $\downarrow \; \downarrow \; \downarrow \; \downarrow$

Positive integers: $1, \; 2, \; 3, \; 4, \; \ldots$

$2n$ corresponds to $n+1$

Example: The set of rational numbers is countable

Rational numbers: $\frac{1}{2}, \frac{3}{4}, \frac{7}{8}, \ldots$

Naïve Approach

Nominator 1

Rational numbers: $\frac{1}{1}, \frac{1}{2}, \frac{1}{3}, \ldots$

Correspondence: $\downarrow \quad \downarrow \quad \downarrow$

Positive integers: $1, 2, 3, \ldots$

Doesn't work:

we will never count numbers with nominator 2: $\frac{2}{1}, \frac{2}{2}, \frac{2}{3}, \ldots$

Better Approach

$$\frac{1}{1} \qquad \frac{1}{2} \qquad \frac{1}{3} \qquad \frac{1}{4} \qquad \cdots$$

$$\frac{2}{1} \qquad \frac{2}{2} \qquad \frac{2}{3} \qquad \cdots$$

$$\frac{3}{1} \qquad \frac{3}{2} \qquad \cdots$$

$$\frac{4}{1} \qquad \cdots$$

$$
\begin{array}{ccccccccc}
\dfrac{1}{1} & \longrightarrow & \dfrac{1}{2} & & \dfrac{1}{3} & & \dfrac{1}{4} & & \cdots \\[2em]
\dfrac{2}{1} & & \dfrac{2}{2} & & \dfrac{2}{3} & & \cdots & & \\[2em]
\dfrac{3}{1} & & \dfrac{3}{2} & & \cdots & & & & \\[2em]
\dfrac{4}{1} & \cdots & & & & & & &
\end{array}
$$

$$
\begin{tikzcd}
\frac{1}{1} \arrow[r] & \frac{1}{2} \arrow[dl] & \frac{1}{3} & \frac{1}{4} & \cdots \\
\frac{2}{1} & \frac{2}{2} & \frac{2}{3} & \cdots & \\
\frac{3}{1} & \frac{3}{2} & \cdots & & \\
\frac{4}{1} & \cdots & & &
\end{tikzcd}
$$

$$
\begin{array}{ccccc}
\frac{1}{1} & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \cdots \\[2em]
\frac{2}{1} & \frac{2}{2} & \frac{2}{3} & \cdots & \\[2em]
\frac{3}{1} & \frac{3}{2} & \cdots & & \\[2em]
\frac{4}{1} & \cdots & & &
\end{array}
$$

$$
\begin{array}{ccccc}
\frac{1}{1} & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \cdots \\
\frac{2}{1} & \frac{2}{2} & \frac{2}{3} & \cdots & \\
\frac{3}{1} & \frac{3}{2} & \cdots & & \\
\frac{4}{1} & \cdots & & &
\end{array}
$$

$$
\begin{array}{ccccc}
\frac{1}{1} & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \cdots \\[2ex]
\frac{2}{1} & \frac{2}{2} & \frac{2}{3} & \cdots & \\[2ex]
\frac{3}{1} & \frac{3}{2} & \cdots & & \\[2ex]
\frac{4}{1} & \cdots & & &
\end{array}
$$

Rational Numbers: $\frac{1}{1}, \frac{1}{2}, \frac{2}{1}, \frac{1}{3}, \frac{2}{2}, \ldots$

Correspondence:

Positive Integers: $1, \ 2, \ 3, \ 4, \ 5, \ \ldots$

We proved:

the set of rational numbers is countable
by describing an enumeration procedure
(enumerator)
for the correspondence to natural numbers

Definition

Let S be a set of strings (Language)

An **enumerator** for S is a Turing Machine that generates (prints on tape) all the strings of S one by one

and

each string is generated in finite time

strings $s_1, s_2, s_3, \ldots \in S$

Enumerator

Machine for S

output

(on tape)

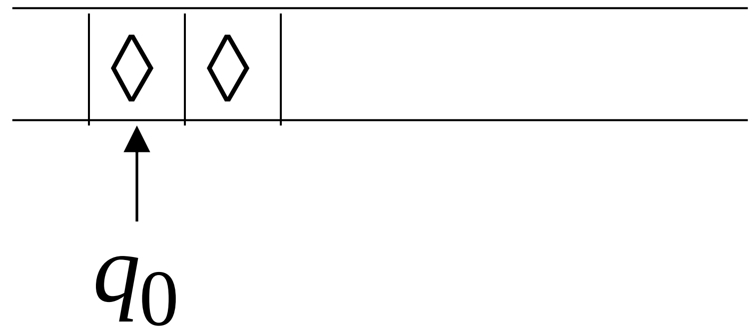
$s_1, s_2, s_3, \ldots$

Finite time: $t_1, t_2, t_3, \ldots$

Enumerator Machine

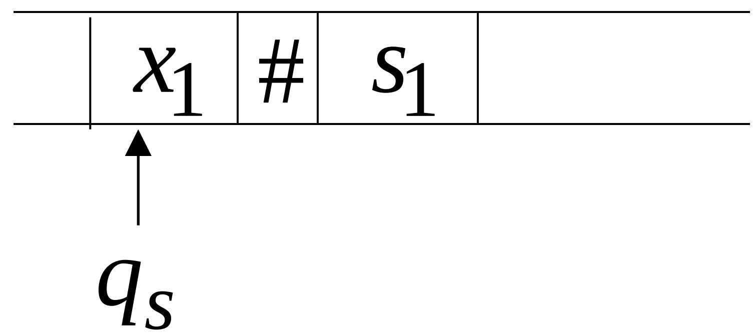
Configuration

Time 0



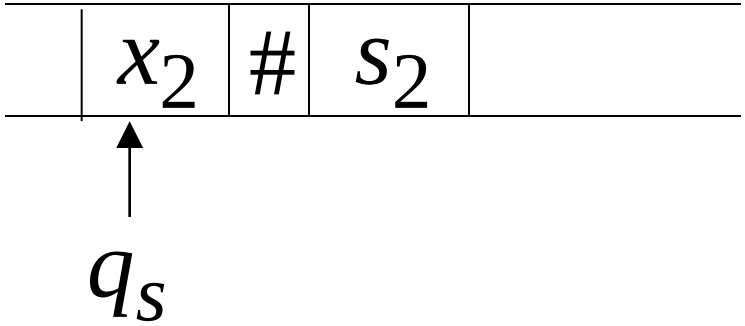
prints s_1

Time t_1



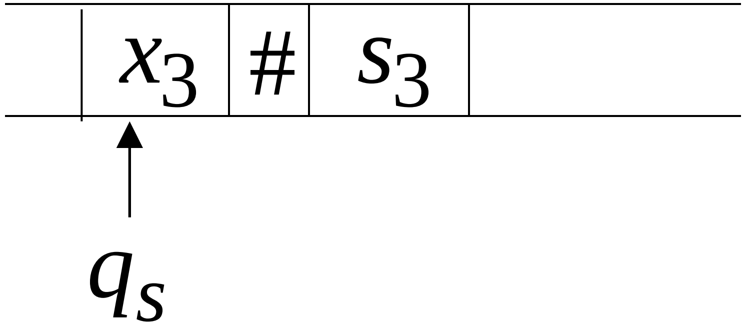
Time t_2

prints s_2



Time t_3

prints s_3



Observation:

If for a set S there is an enumerator, then the set is countable

The enumerator describes the correspondence of S to natural numbers

Example: The set of strings $S = \{a, b, c\}^{+}$ is countable

Approach:

We will describe an enumerator for S

Naive enumerator:

Produce the strings in lexicographic order:

$$
\begin{aligned}
s_1 &= a \\
s_2 &= aa \\
&\vdots \quad aaa \\
&\quad\; aaaa \\
&\quad\; \ldots\ldots
\end{aligned}
$$

Doesn't work:

strings starting with b
will never be produced

Better procedure: **Proper Order**

(Canonical Order)

1. Produce all strings of length 1

2. Produce all strings of length 2

3. Produce all strings of length 3

4. Produce all strings of length 4

....

Produce strings in

Proper Order:

$$\left.\begin{array}{l} s_1 = a \\ s_2 = b \\ \vdots \quad c \end{array}\right\} \text{length 1}$$

$$\left.\begin{array}{l} aa \\ ab \\ ac \\ ba \\ bb \\ bc \\ ca \\ cb \\ cc \end{array}\right\} \text{length 2}$$

$$\left.\begin{array}{l} aaa \\ aab \\ aac \\ \ldots\ldots \end{array}\right\} \text{length 3}$$

Theorem: The set of all Turing Machines is countable

Proof: Any Turing Machine can be encoded with a binary string of 0's and 1's

Find an enumeration procedure for the set of Turing Machine strings

Enumerator:

Repeat

1. Generate the next binary string of 0's and 1's in proper order

2. Check if the string describes a Turing Machine

   if **YES:** print string on output tape

   if **NO:** ignore string

Binary strings		Turing Machines
0	
1	
00	
01	
⋮	
1 0 1 0 11 0 11 0 0	
1 0 1 0 11 0 11 0 1	$\xrightarrow{s_1}$	1 0 1 0 11 0 11 0 1
⋮	
1 0 11 0 1010010101101	$\xrightarrow{s_2}$	1 0 11 0 1010010101101
⋮	

End of Proof

Uncountable Sets

We will prove that there is a language L' which is not accepted by any Turing machine

Technique:

Turing machines are countable

Languages are uncountable

(there are more languages than Turing Machines)

Definition: A set is uncountable if it is not countable

We will prove that there is a language which is not accepted by any Turing machine

Theorem:

If S is an infinite countable set, then

the powerset 2^S of S is uncountable.

(the powerset 2^S is the set whose elements are all possible sets made from the elements of S)

Proof:

Since S is countable, we can write

$$S = \{s_1, s_2, s_3, \ldots\}$$

Elements of S

Elements of the powerset 2^S have the form:

$$\emptyset$$

$$\{s_1, s_3\}$$

$$\{s_5, s_7, s_9, s_{10}\}$$

......

We encode each element of the powerset with a binary string of 0's and 1's

Powerset element (in arbitrary order)	Binary encoding s_1	s_2	s_3	s_4	$\cdots$
$\{s_1\}$	1	0	0	0	$\cdots$
$\{s_2, s_3\}$	0	1	1	0	$\cdots$
$\{s_1, s_3, s_4\}$	1	0	1	1	$\cdots$

Observation:

Every infinite binary string corresponds to an element of the powerset:

Example: $1\,0\,0\,1\,1\,1\,0\quad\cdots$

Corresponds to: $\{s_1, s_4, s_5, s_6, \ldots\} \in 2^S$

Let's assume (for contradiction)
that the powerset 2^S is countable

Then: we can enumerate
the elements of the powerset

$$2^S = \{t_1, t_2, t_3, \ldots\}$$

Powerset element	suppose that this is the respective **Binary encoding**					
t_1	1	0	0	0	0	$\cdots$
t_2	1	1	0	0	0	$\cdots$
t_3	1	1	0	1	0	$\cdots$
t_4	1	1	0	0	1	$\cdots$
$\cdots$			$\cdots$			

Take the binary string whose bits are the complement of the diagonal

t_1	1	0	0	0	0	$\cdots$
t_2	1	1	0	0	0	$\cdots$
t_3	1	1	0	1	0	$\cdots$
t_4	1	1	0	0	1	$\cdots$

Binary string: $\boldsymbol{t} = 0011\ldots$

(birary complement of diagonal)

The binary string $\quad t = 0011\ldots$

corresponds
to an element of
the powerset 2^S: $\quad t = \{s_3, s_4, \ldots\} \in 2^S$

Thus, t must be equal to some t_i

$$t = t_i$$

However,

the i-th bit in the encoding of t is
the complement of the i-th bit of t_i, thus:

$$t \neq t_i$$

Contradiction!!!

Since we have a contradiction:

The powerset 2^S of S is uncountable

End of proof

An Application: Languages

Consider Alphabet : $A = \{a, b\}$

The set of all Strings:

$$S = \{a, b\}^* = \{\lambda, a, b, aa, ab, ba, bb, aaa, aab, \ldots\}$$

infinite and countable

(we can enumerate the strings
in proper order)

Consider Alphabet : $A = \{a, b\}$

The set of all Strings:

$$S = \{a, b\}^* = \{\lambda, a, b, aa, ab, ba, bb, aaa, aab, \ldots\}$$

infinite and countable

Any language is a subset of S :

$$L = \{aa, ab, aab\}$$

Consider Alphabet : $A = \{a, b\}$

The set of all Strings:

$$S = A^* = \{a, b\}^* = \{\lambda, a, b, aa, ab, ba, bb, aaa, aab, \ldots\}$$

infinite and countable

The powerset of S contains all languages:

$$2^S = \{\emptyset, \{\lambda\}, \{a\}, \{a, b\}, \{aa, b\}, \ldots, \{aa, ab, aab\}, \ldots\}$$

uncountable

Consider Alphabet : $A = \{a, b\}$

countable

Turing machines: $M_1 \quad M_2 \quad M_3 \quad \cdots$

accepts

Languages accepted By Turing Machines: $L_1 \quad L_2 \quad L_3 \quad \cdots$

countable

Denote: $X = \{L_1, L_2, L_3, \ldots\}$

countable

Note: $X \subseteq 2^S$

$(S = \{a, b\}^*)$

Languages accepted by Turing machines: X countable

All possible languages: 2^S uncountable

Therefore: $X \neq 2^S$

$$\left(\text{since } X \subseteq 2^S, \text{ we have } X \subset 2^S\right)$$

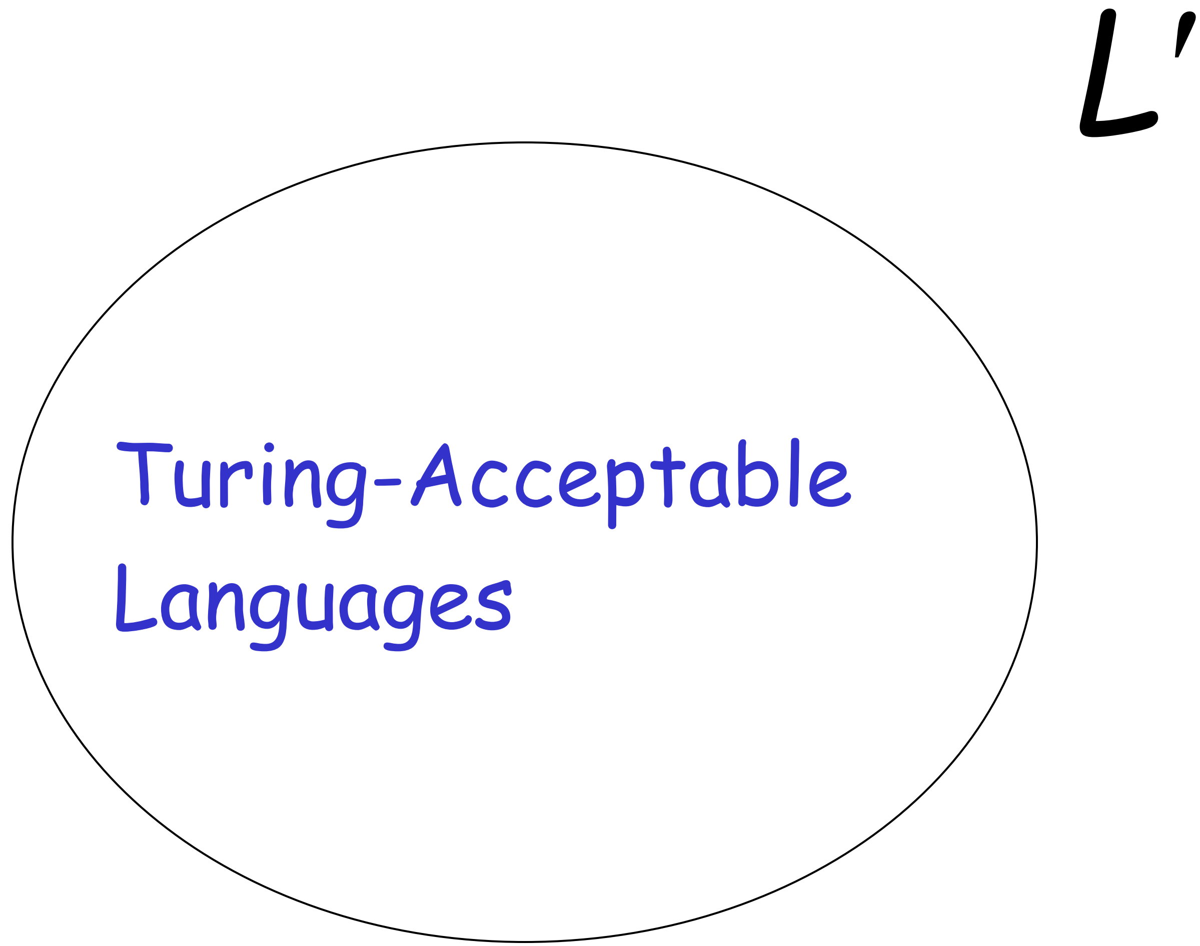
Conclusion:

There is a language L' not accepted by any Turing Machine:

$$X \subset 2^S \Longrightarrow \exists L' \in 2^S \text{ and } L' \notin X$$

(Language L' cannot be described by any algorithm)

Non Turing-Acceptable Languages



Note that: $X = \{L_1, L_2, L_3, \ldots\}$

is a *multi-set* (elements may repeat) since a language may be accepted by more than one Turing machine

However, if we remove the repeated elements, the resulting set is again countable since every element still corresponds to a positive integer